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1 % The wrench must sustain a fully reversed torque of  $T = \pm 600$  in-lbf for  $10^6$  cycles.
   Design
2 % will include selecting an appropriate material and dimensions to meet or exceed the
   following
3 % requirements:
4
5 Xo_req = 4; %safety factor against yield or brittle failure
6 Xk_req = 2; %safety factor for crack growth from assumed crack of depth 0.04 in
7 Xs_req = 1.5; %fatigue stress safety factor
8 min_output = 1; %mV/V
9
10 % Load, dimensions and mechanical properties:
11 M = 600; % max torque, in-lbf
12 L = 16; % length from drive to where load applied, in
13 h = 0.7; % width, in
14 b = 0.4; % thickness, in
15 I = (b*h^3)/12; %moment of inertia, in^4
16 c = 1.0; % distance from center of drive to center of strain gauge, in
17 E = 16.1*10^6; % Young's modulus, psi
18 nu = 0.36; % Poisson's ratio
19 su = 148*10^3; % yield strength, psi
20 KIC = 74.6 * 10^3; % fracture toughness, psi sqrt(in)
21 sfatigue = 87.4 * 10^3; % fatigue strength from Granta for  $10^6$  cycles, psi
22 name = 'Ti-6Al-4V, aged'; % material name, must be steel, aluminum, or titanium alloy
23 k = 2; %gauge calibration factor
24
25 % stress and deflection analysis
26 umax = (M*L^2)/(3*E*I); %maximum deflection in the positive y direction, in
27 max_normal_stress = 6*M/(b*h^2); %psi
28 M_gauge = (M*(L-c))/L;
29 gauge_stress = 6*M_gauge/(b*h^2);
30 required_force = M/L;
31
32 % find safety factors
33 Xo = su/max_normal_stress;
34 a = 0.04; % assumed crack depth, in
35 KI = 1.12*max_normal_stress*sqrt(pi*a);
36 Xk = KIC/KI;
37 Xs = sfatigue/max_normal_stress;
38
39 % strain gauge results:
40 gauge_strain = gauge_stress/E;
41 output = 0.5 * gauge_strain * k * 1000; %mV/V at 600 in-lbf using half bridge
42
43 %results
44 fprintf('load point deflection: %.3f in\n', umax);
45 fprintf('user input force: %.3f lbs\n', required_force);
46 fprintf('max normal stress: %.2f ksi\n', max_normal_stress/1e3);
47 fprintf('normal stress at gauge: %.2f psi\n', gauge_stress);
48 fprintf('safety factor for strength: %.1f\n', Xo);
49 if Xo < Xo_req
50     fprintf('FAIL: Xo is too low');
51 end
52 fprintf('safety factor for crack growth: %.2f\n', Xk);
53 if Xk < Xk_req
54     fprintf('FAIL: Xk is too low');
55 end
56 fprintf('safety factor for fatigue: %.2f\n', Xs);
57 if Xs < Xs_req
58     fprintf('FAIL: Xs is too low');
59 end

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60 fprintf('strain at gauge: %.2f microstrain\n', gauge_strain * 10^6);
61 fprintf('bridge output: %.3f mV/V\n', output);
62 if output < min_output
63     fprintf('FAIL: output is too low');
64 end
65
66 if output >= min_output && Xo >= Xo_req && ...
67     Xs >= Xs_req && Xk >= Xk_req
68     fprintf('\nDesign meets all requirements.\n');
69 else
70     fprintf('\nDESIGN DOES NOT MEET REQUIREMENTS.\n');
71 end
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